Bone-Crete

Applying the inner matrix of bone to concrete

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APPL101 – 001

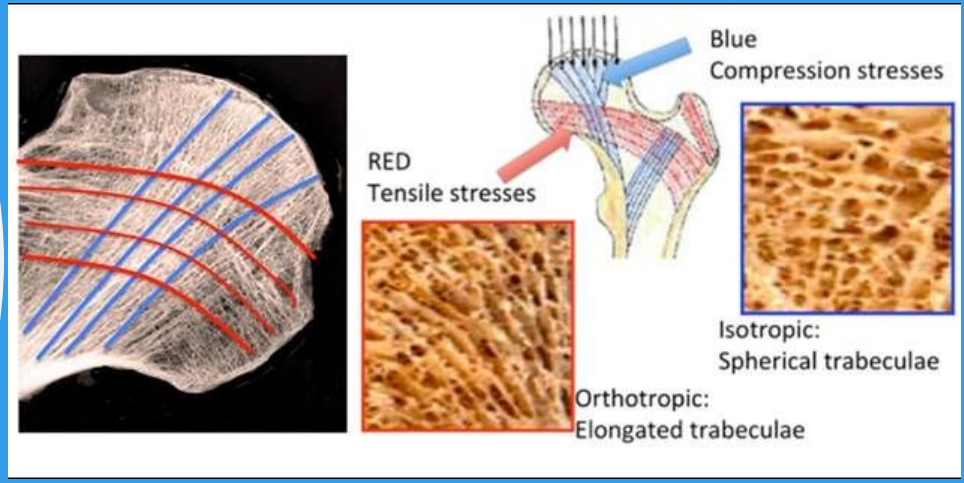


Project Overview

- Traditional concrete is **heavy**, **material-intensive**, and contributes to 8% of global CO_2 emissions.
- Inspired by the **efficiency of bone** structures.
- Addresses the urgent need for **sustainable construction materials**.
- Reimagines the future of the construction industry.

Biomimicry Inspiration

- Bones are not completely solid; durability in concrete is typically inversely proportional to porosity, but strategically distributing “bubbles” to create a trabecular matrix can have the opposite effect.
- Shapes in real bones are very precise, but this is acceptable in concrete due to molding and casting techniques.



Force distribution across trabeculae
Aversa et al., *Trabecular Prostheses* 9/19/2019

Solution

- **Traditional concrete:** Heavy, inefficient, unsustainable due to greater material usage
- **Bone-inspired design:** Efficient and strong with less materials
- Prototypes test feasibility for real-world use



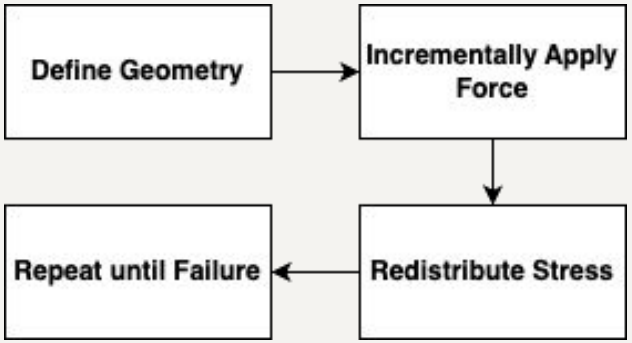
Traditional Concrete



Proof of Concept

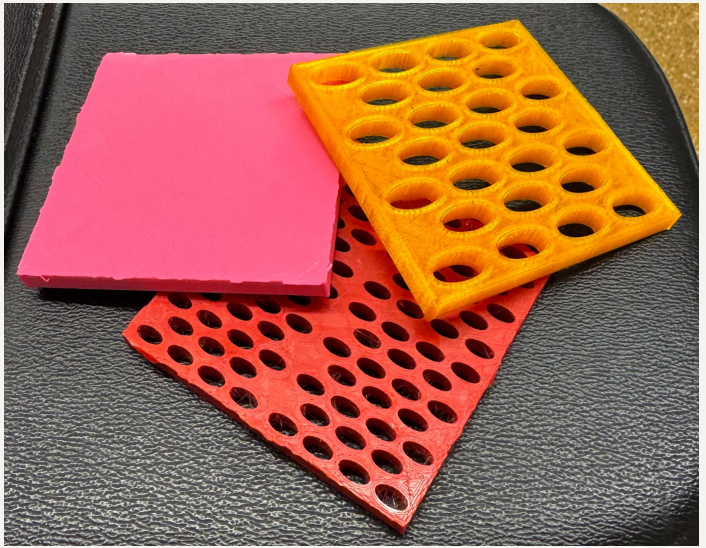
Optimization

Modeling

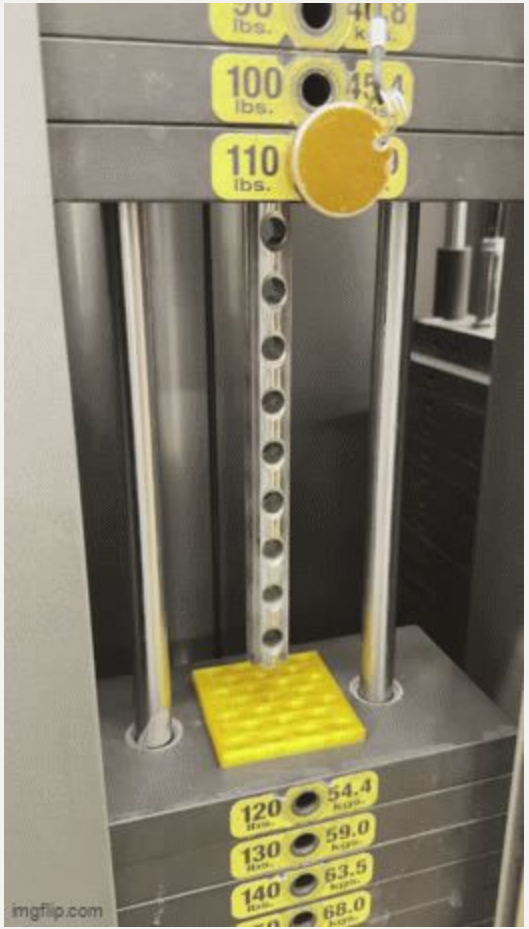


Prototypes

3D Printed (PLA), 3x3x0.25 inches

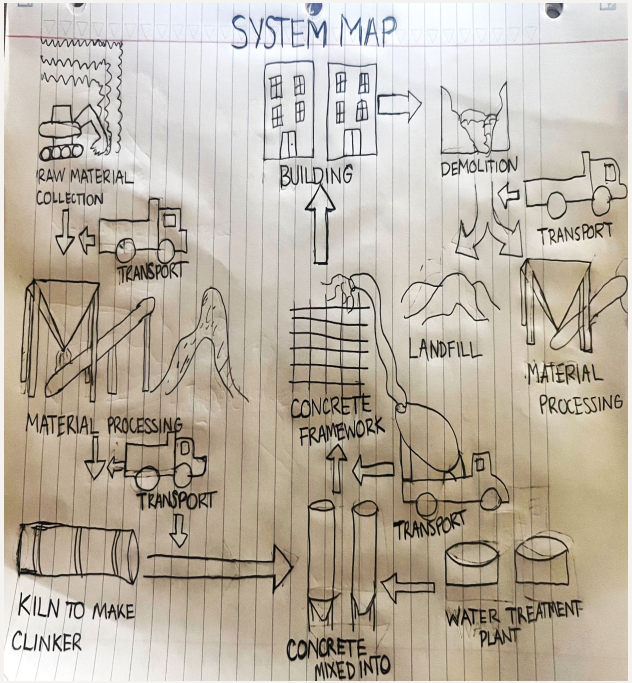



150 lbs.
160 lbs. 72.6 kgs.
170 lbs. 77.1 kgs.
180 lbs. 81.6 kgs.
190 lbs. 86.2 kgs.
200 lbs. 90.7 kgs.
imgflip.com


100 lbs.
110 lbs.
120 lbs. 54.4 kgs.
130 lbs. 59.0 kgs.
140 lbs. 63.5 kgs.
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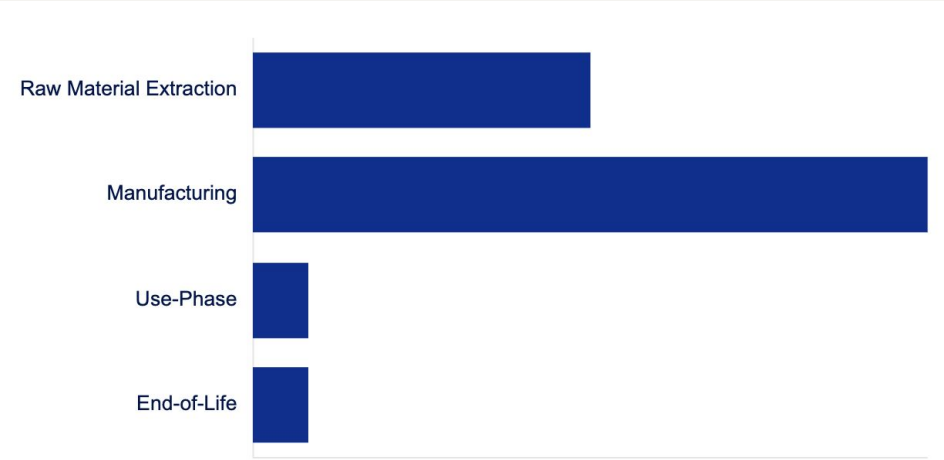

90 lbs.
100 lbs.
110 lbs.
130 lbs. 59.0 kgs.
140 lbs. 63.5 kgs.

Sustainability Analysis



Systems Map

Life Cycle Analysis



Design Influence

- Raw material extraction and manufacturing have greatest impact on sustainability.
- Maintain durability.
- Decrease material usage.

The Three P's

People

- Consumers and producers
- Ensure safety standards
- Preserve Employment

Profit

- Financial Viability.
- Minimal Disruption
- Infrastructure

Planet

- CO_2 emissions
- Habitat destruction
- Raw material depletion

People vs. Profit

- Profit & wellness
- Increased margins
- Sustainability & public health

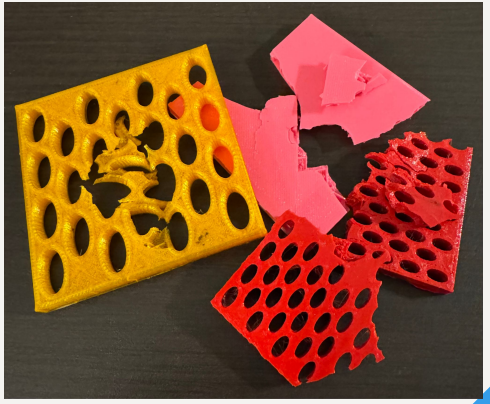
Profit vs. Planet

- Profit & sustainability
- Lower emissions.
- Waste reduction
- Industry incentive

People vs. Planet

- Safety & Sustainability
- Worker & Local Businesses
- Preserving industry framework

Discussion



Experimental Results:

- **Small-hole model:** Greater tensile strength, withstood higher forces before failure.
- **Large-hole model:** Localized impact better, reducing fracture to a smaller region.

Real-World Applications:

- **Small-hole design:** Prioritizes overall strength and load resistance.
- **Large-hole design:** Improves durability under uneven loading or impact scenarios.

Questions?

